

Linear Algebra I

beginning: polynomials

Linear: the equation is of degree one.

$$aX + bY = c.$$

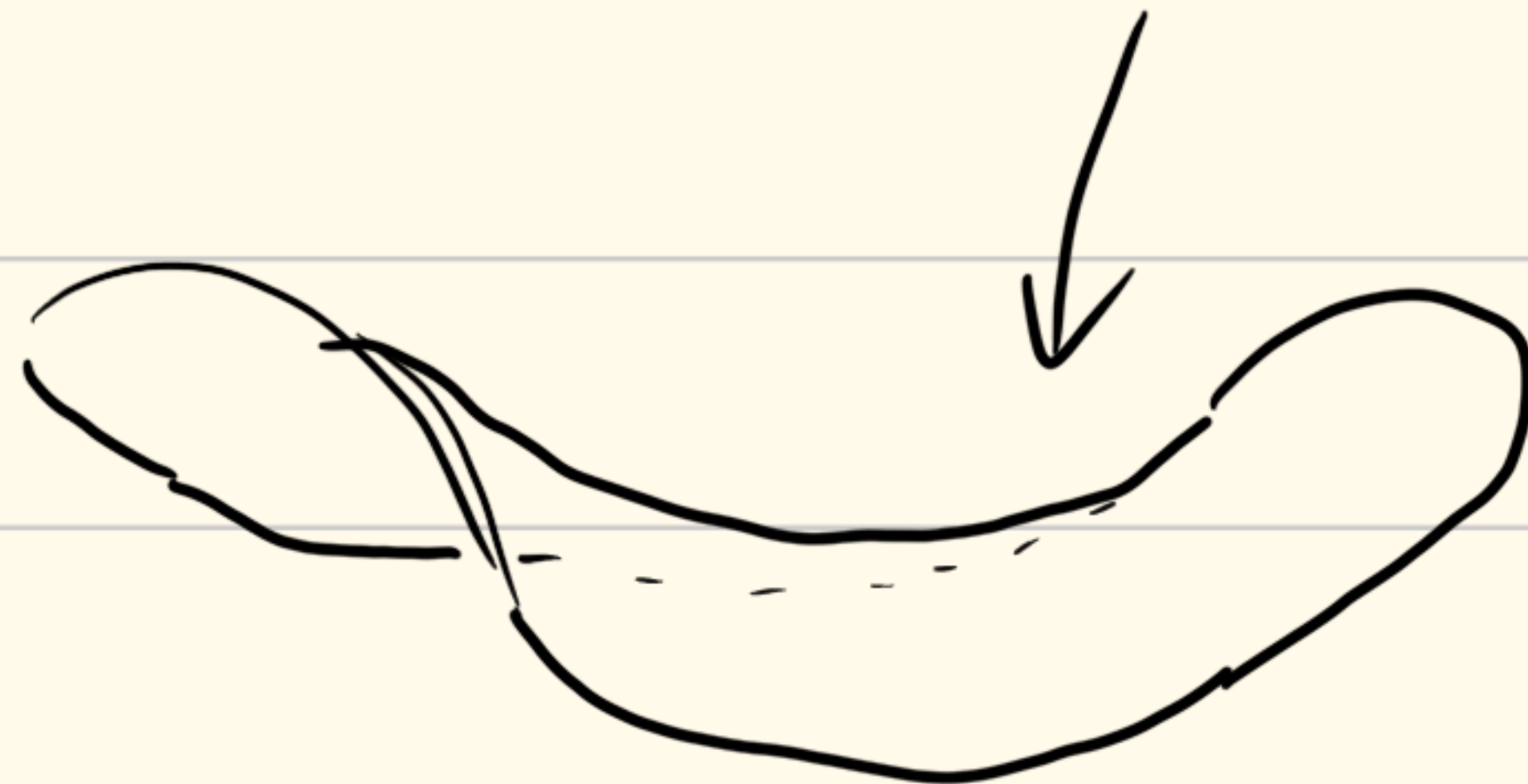
You won't see $X^2 + 3XY + Y^4 = 9$.

such equations. Analysis or AG.

mathematics ? hard to define. $x^n + y^n = z^n$, integer solutions

- number theory (integer solutions) Fermat's last theorem. 5次方程没有根式解
- algebra (polynomials) FTA $\mathbb{C}$ -coefficients degree n polynomial
- analysis (calculus) FTC Newton-Leibniz formula.
- topology (topological invariants) $\downarrow$ under continuous deformation convex polyhedron. $V - E + F = 2$. Knot theory.
- geometry (space) sum of inner angles $\neq 180^\circ$.
- probability theory Riemann Geometry.
- combinatorics Hyperbolic Geometry





$$\begin{cases} x+y+z=6. \\ 2x+3y+z=11. \\ x+2y+3z=14. \end{cases}$$

Linear Algebra is everywhere.

显卡 Graphics. 矩阵计算. CS2.

解方程, solve systems of degree one equations.

1.10.

$$\boxed{Ax=b} \stackrel{??}{\iff} Ax=0$$

nonhomogeneous

homogeneous.

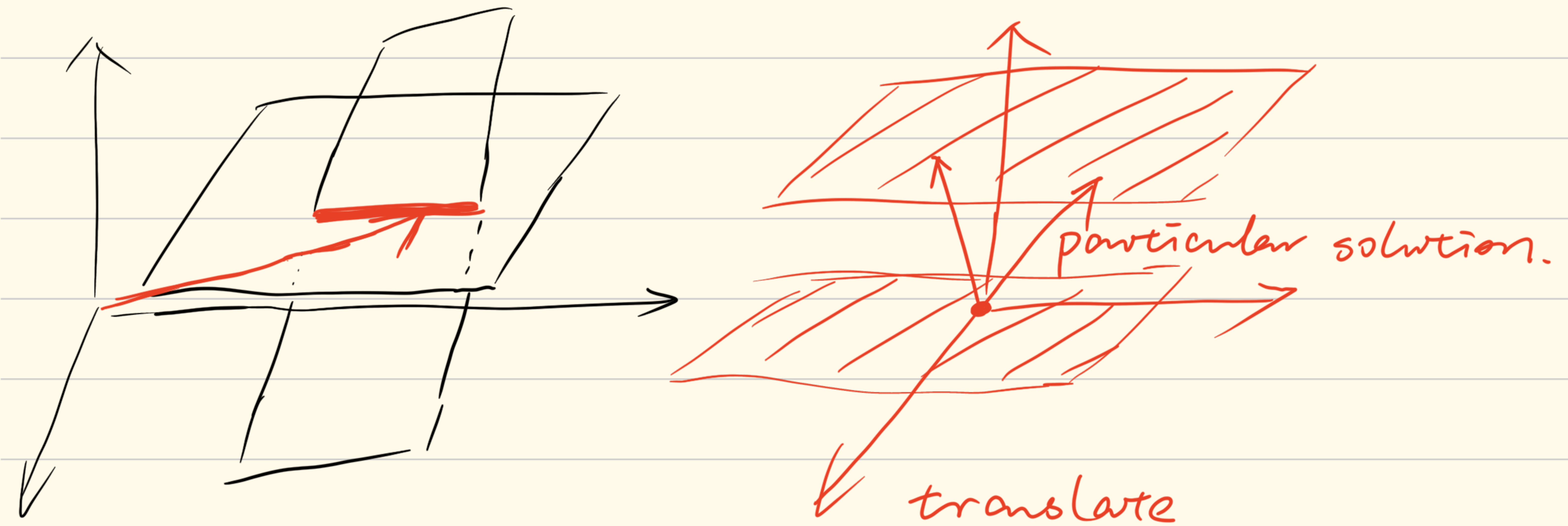
solution

General solution of $Ax=b$ = One particular + general solution of $Ax=0$.

Assume $Av=0$. $A(x_0+v) = Ax_0 + Av = Ax$.

If x_0 solves $Ax_0=b$, so does x_0+v .

$$\forall Ax = b, A(x - x_0) = Ax - Ax_0 = b - b = 0. \quad \square$$



independent from the choice of the particular solution.

1.14 (2)

$$\begin{cases} \lambda x_1 + x_2 + x_3 = 1. \\ x_1 + \lambda x_2 + x_3 = 1. \\ x_1 + x_2 + \lambda x_3 = 1. \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & \lambda & 1 & 1 \\ \lambda & 1 & 1 & 1 \\ 1 & 1 & \lambda & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & \lambda & 1 & 1 \\ 0 & 1-\lambda^2 & 1-\lambda & 1-\lambda \\ 0 & 1-\lambda & \lambda-1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & \lambda & 1 & 1 \\ 0 & 1-\lambda & \lambda-1 & 1-\lambda \\ 0 & 0 & (1-\lambda)(\lambda+2) & 0 \end{array} \right]$$

$$\left[\begin{array}{c} 1 \\ 0 \\ 1-\lambda \end{array} \right] \quad \text{I}^\circ. \lambda = 1, \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + S \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$2^\circ \lambda \neq 1 \quad \left[\begin{array}{ccc|c} 1 & \lambda & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & \lambda+2 & 1 \end{array} \right]$$

If $\lambda = -2$, no solution. If $\lambda \neq -2$, $x_1 = x_2 = x_3 = \frac{1}{\lambda+2}$.

Alternate solution: $(\lambda-1)(x_1-x_2)=0$, $(\lambda-1)(x_2-x_3)=0$.

$\lambda = 1 \checkmark$ $\lambda \neq 1$, $x_1 = x_2 = x_3$, $\lambda \neq -2$, or $\lambda = 2$.

(选讲)

distinct

1.15 (2) $P_i = (x_i, y_i)$ $x^2 + y^2 + Dx + Ey + F = 0$.

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \\ x_4 & y_4 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ x_3^2 + y_3^2 \\ x_4^2 + y_4^2 \end{pmatrix}$$

$Ax = b$ 有解?

$\leadsto (D, E, F)$

$$\left(x + \frac{D}{2}\right)^2 + \left(y + \frac{E}{2}\right)^2 = \frac{D^2 + E^2}{4} - F.$$

有没有可能有无穷个解? (三点确定一个(直线)圆).

> 0 .

Is there a linear system (with real variable) with exactly two solutions?

Solutions? (infinite, unique, no solution).

$$Ax = b, A \in \mathbb{R}^{m \times n}$$

$$[0 \ 0 \ \dots \ 0 \mid b \neq 0]$$

1. 高斯消元法 (augmented matrix) n variables rank r
 $n-r$ free variables.

Hard to say, $m \neq n$. $\text{rank } A \neq n$ still hard.

Assume $m = n$. $Ax = 0$ only zero solution $\Leftrightarrow \text{rank } A = n$.

$\Leftrightarrow Ax = b$ has unique solution for all $b \in \mathbb{R}^n$.

$\Leftrightarrow Ax = b$ has unique solution for some $b \in \mathbb{R}^n$.

(1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1).

